

Gazipur Digital University

Department of Software Engineering

System Requirements Specification (SRS)

for

“ReVendo: Buy, Sell & Repair Pre-Owned Items”

Submitted By:

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Step 1: Introduction

This document specifies the Software Requirements Specification (SRS) for **ReVendo: Buy, Sell & Repair Pre-Owned Items**. It defines the functional and non-functional requirements of the system and provides a reference for all stakeholders involved in the development process. This SRS covers the requirements for **the entire ReVendo system**, including the web/mobile marketplace platform, integrated repair services, communication system, environmental impact tracking, and administrative management tools. The software is a self-contained system with optional integrations to external payment gateways and notification services. This document is intended for developers, project managers, testers, system administrators, business stakeholders, repair service partners, and end users.

Project Scope

Product Description: ReVendo is a web and mobile application platform designed to facilitate secure second-hand transactions while promoting environmental sustainability through integrated repair services and circular economy principles in Bangladesh. The system automates marketplace operations, reduces transaction fraud, and extends product lifecycles through coordinated repair and reuse initiatives.

Relation to Business Goals: This software supports strategic objectives including capturing the growing second-hand market in Bangladesh's USD 6.5 billion B2C e-commerce sector, demonstrating environmental leadership through sustainable technology solutions, and creating economic opportunities for repair service providers while reducing electronic waste by up to 70% compared to new product purchases.

Step 2: Overall Description

1. Product Perspective

ReVendo is a new, self-contained system designed to facilitate secure second-hand transactions while promoting environmental sustainability in Bangladesh. The system resolves limitations of existing informal trading methods (social media groups) including inefficiency, lack of transparency, and fraud risks. ReVendo interacts with external systems such as payment gateways, SMS/email services, and local repair service networks.

2. Product Features

- **User Registration and Authentication:** Secure account creation and login
- **Product Marketplace Management:** Listing, browsing, and purchasing pre-owned items
- **Enhanced Communication System:** Real-time messaging between users

- **Environmental Impact Tracking:** Eco-score calculation based on user activities
- **Administrative Management:** Platform oversight and moderation tools
- **User Activity History Management:** Transaction history tracking

3. User Classes and Characteristics

(a) Registered Users (Buyers/Sellers/Donors)

- Frequency: Daily to weekly
- Functions: List/buy products, donate, request repairs, messaging
- Expertise: Low to moderate technical knowledge
- Privileges: Own products and transactions only

(b) Repair Service Partners

- Frequency: Daily
- Functions: Manage repair requests, update status, customer communication
- Expertise: Moderate technical + specialized repair skills
- Privileges: Repair request management

(c) System Administrators

- Frequency: Daily
- Functions: User verification, content moderation, system configuration
- Expertise: High technical knowledge
- Privileges: Full administrative access

4. Operating Environment

Hardware Platform: Android/iOS smartphones, desktop computers, cloud servers **Operating System:** Android 8+, iOS 12+, Linux servers **Software Dependencies:** Web frameworks, mobile frameworks, WebSocket, cloud storage **Other Requirements:** Intermittent connectivity support, low-end device compatibility

5. Design and Implementation Constraints

- Must comply with Bangladesh data protection and e-commerce regulations
- Mandatory HTTPS/TLS encryption and secure password storage
- Cross-platform compatibility requirements
- Bandwidth optimization for slow internet connections
- Integration with local repair service networks
- Cost-effective infrastructure within project budget

Step 3: System Features

Feature 1: User Registration & Authentication

- **Description:** Secure account creation and login for marketplace access.
- **Priority:** High (Benefit: 9, Penalty: 9, Cost: 4, Risk: 8)
- **Requirements:**
 - Register with unique username, password, phone, address
 - Authenticate via username/password
 - Prevent duplicate usernames
 - Lock account after 3 failed attempts
 - Store hashed passwords
 - Support sessions & logout

Feature 2: Product Marketplace Management

- **Description:** Core marketplace for creating, browsing, and purchasing product listings.
- **Priority:** High (Benefit: 9, Penalty: 9, Cost: 6, Risk: 6)
- **Requirements:**
 - Create and display product listings with filters/search
 - Buyers can purchase available items
 - Remove purchased items from listings
 - Record transactions for buyer & seller
 - Provide seller contact info post-purchase

Feature 3: Communication System

- **Description:** Real-time messaging and notifications between users.
- **Priority:** High (Benefit: 8, Penalty: 7, Cost: 5, Risk: 5)
- **Requirements:**
 - One-to-one chat with timestamps
 - Persist and sort conversations
 - Unread/read message tracking
 - Automated notifications
 - Search users to start chats

Feature 4: Environmental Impact Tracking

- **Description:** Tracks user eco-score and carbon savings from sustainable actions.
- **Priority:** Medium-High (Benefit: 7, Penalty: 5, Cost: 4, Risk: 3)
- **Requirements:**
 - Calculate eco-scores and carbon saved
 - Display activity breakdown (donated, repaired, sold, bought)
 - Configurable impact weights (admin)
 - Track cumulative impact

Feature 5: Administrative Management

- **Description:** Admin tools for user, content, and system oversight.
- **Priority:** High (Benefit: 9, Penalty: 9, Cost: 6, Risk: 8)
- **Requirements:**
 - Secure admin login & dashboard
 - Manage users and product listings
 - Chat audits and content moderation
 - Suspend/ban users
 - Log all admin actions

Feature 6: User Activity History

- **Description:** Maintains records of user transactions and activities.
- **Priority:** Medium (Benefit: 6, Penalty: 4, Cost: 3, Risk: 2)
- **Requirements:**
 - Record all user transactions automatically
 - Categorize by type (buy, sell, donate, repair)
 - Show item details and chronology
 - Provide complete personal history

Step 4: External Interface Requirements

(a) User Interfaces

- **Logical Design:** The system provides a text-based, menu-driven console interface. Users interact with numbered menus and prompts.
- **Components Requiring UIs:** Registration/Login, User Dashboard (product actions, chat, eco-impact, history), and Admin Panel.
- **GUI Standards:** Consistent layout and navigation across all menus.
- **Error Message Conventions:** Standardized feedback such as *“Invalid choice”*, *“Invalid credentials”*, and *“Username already exists”*.
- **Accessibility Considerations:** Lightweight console design ensures compatibility across operating systems and hardware platforms.

(b) Hardware Interfaces

- The system operates on standard desktop or laptop hardware with a Java Runtime Environment.
- No external sensors or devices are required.
- Data is stored in local files (`users.dat`, `products.dat`, `chats.dat`) on the host machine.

(c) Software Interfaces

- **Operating System:** Cross-platform (Windows, Linux, macOS) via Java Virtual Machine (JVM).
- **Databases:** File-based persistence using Java serialization (ObjectInputStream/ObjectOutputStream).
- **Libraries and Tools:** Standard Java libraries (java.io, java.util, java.text).
- **Data Exchange:** Internal object serialization for persistent storage; no external API integration in the current version.

(d) Communications Interfaces

- The system currently functions as a **standalone application** without internet or network dependencies.
- Communication between users is supported via an **internal chat system** stored locally.
- Future versions may extend to real-time communication through sockets or REST-based APIs.

(e) Nonfunctional Requirements

- **Performance Requirements:** Data loading and saving shall complete within two (2) seconds. The system shall support up to 1,000 users and 5,000 products with acceptable performance.
- **Safety Requirements:** The system shall handle invalid input gracefully and ensure persistence of data through controlled serialization to prevent corruption.
- **Security Requirements:** Usernames shall be unique. Admin authentication shall be restricted via credentials. Passwords are currently stored in plain text; future releases shall adopt secure hashing.
- **Software Quality Attributes:**
 - *Portability:* Runs on any operating system with a JVM.
 - *Usability:* Clear console menus and intuitive navigation.
 - *Maintainability:* Modular class design (User, Product, Chat, ChatManager).
 - *Reliability:* Data is auto-saved after key operations to reduce risk of loss.
 - *Scalability:* Current file-based storage is suitable for small-scale use; migration to a database is recommended for larger deployments.

Step 5: Other Requirements

- **Database:** Local file-based storage (users.dat, products.dat, chats.dat); future upgrade to SQL/Cloud DB.
- **Internationalization:** English supported; future option for Bangla/multi-language.
- **Legal/Ethical:** Comply with consumer protection laws; protect user data; include product liability disclaimers.
- **Reuse/Extensibility:** Modular design for reuse in mobile/web apps; future integration with payment gateways and repair services.

- **Backup/Recovery:** Data auto-saved after actions; reload previous state on restart.
- **Future Enhancements:** Secure payments, cloud storage, mobile app, AI-based recommendation.